Module 2 — Read 0

Project preview — Project 2 — Estimating the Truth

Module 2 Preview

Big question: What is the truth about a whole population, and how close can we get from one sample?

> **Note**
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> This page is a short orientation. Read it before Read 1 so you see how Project 2 fits the course story (five steps, three worlds, and plan before you collect).

Good science: plan your estimate before you collect

In Module 0, Read 0 you met the **five-step statistical process**. Module 1 used it to **test a claim** with a p-value. Project 2 uses it to **estimate a parameter** from limited information (a *sample*).

Step	In Project 2
1 — Ask	A research question that asks "What is the value of a **population parameter?**". Your **parameter** should be a **proportion** p or **mean** μ for a **population** that can be listed and sampled.
2 — Plan	State the population (N), the variable, the sampling method (you'll be required to use the simple random sampling method), sample size n, and which **95% Confidence Interval** you will use (mean vs proportion; FPC or not) before you collect data.
3 — Collect	Randomly select n units from a list of the population; visit or inspect each of these units and record the value of your variable once for each unit.
4 — Analyze	Compute a single best estimate ($\hat{p}$ or $\bar{x}$), graph the data, check interval conditions and then compute the **95% Confidence Interval** in R according to your plan.
5 — Conclude	Interpret the interval in context *"I am 95% confident that…"* for your population parameter of interest from the research question.

Statistics moves among three worlds (Module 0). In this project, your research question is about the value of a summary of a large population but due to constraints of time and money, you won't be able to observe each member of this population. So instead you'll observe only a few units and then compute a range of best guesses for the value in the population.

1. The world we imagine (theory first)

Here are some key ideas and terms in our imaginary world:

- Name a **population** and the **parameter** p or μ you want to learn about.
- Plan **simple random sampling** without replacement: every subset of size n from the frame should be equally likely.
- Choose a **confidence level** (95%) and the **interval method** that was built by using mathematics to go from a sample to a population.

2. The world we observe (observations second)

The world we observe will be representative of the world we imagine if we use a good **sampling method**:

- Visit n units in a randomly selected subset of the population and record the value of the variable for each one.
- Compute $\hat{p} = x/n$ if the variable is binary or $\bar{x}$ if the variable is numerical.
- **Descriptive** graphs (histogram, dot plot, or bar chart) show what you measured: they do not by themselves estimate the population parameter with high confidence.

3. The world of computers (connect the two)

R moves between what you imagine and what you observe:

- `prop.test()` or `binom.test()` — 95% CI for a **proportion** (check 4 BRP conditions and large-count conditions first).
- `t.test()` — 95% CI for a **mean** (check random sample and normality or $n > 30$).
- Multiply the standard error by $\sqrt{(N-n)/(N-1)}$ when the **FPC** applies.
- `ggplot2` — display the sample you observed.

> **Notation preview**
>
> Greek letters (p, μ, σ) usually denote **parameters**, which are population values that are usually **unobserved**. **Statistics** from your sample use Roman/Latin style ($\hat{p}$, $\bar{x}$, s, n), these numbers are **observed** from data. If you measure **every** unit in the population (**census**, $n = N$), you know the parameter and the methods from this module are not needed.

Connecting Observations with Theory

Once you have observed data, we ask:

> What range of values for the **parameter** is plausible, given this sample and the way I sampled?

That range is a **95% confidence interval**.

- A narrow interval means your estimate is precise and more useful.
- A wide interval means more uncertainty about the value of the parameter and is less useful.

> **Key Interpretation**
>
> A **95% confidence interval** means: if we repeated sampling the same way many times, about **95%** of those intervals would contain the true parameter.

Connection with the Rest of the Course

Module 1 used the **exact binomial** distribution to conduct a test of a pair of hypotheses. Module 2 adds **approximate** distribution methods for estimation and a new type of data, **numerical**.

At the end of this module you'll be able to compute a confidence interval in one line of R code AND you'll have a deep understanding of the foundations that will allow you to chose **which R function to use**, **which conditions to check**, and **how to interpret** the output.

Interval formulas (preview)

Proportion (Wald, at least 10 successes and 10 failures):

$$\hat{p} \pm 2\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

(use `prop.test`, apply **FPC** when sampling without replacement from finite N.)

Mean (large n or normal population, when conditions hold):

$$\bar{x} \pm 2\frac{s}{\sqrt{n}}$$

(use `t.test`, apply **FPC** when sampling without replacement from a finite N.)

Tour the sample project

Here is Rosanna's **sample one-page write-up — CA high schools (PDF)**

Section	What to notice	World
Introduction	Research question with a finite population and a parameter (p or μ)	Imagine
Methods	Simple Random Sample (SRS) from $\{1, \ldots, N\}$ of size n from the population of size N. Variable collected.	Observe
Results	Single best guess for the real answer and an interval I am highly confident contains the real answer	Compute
Conclusion	*"I am 95% confident that…"*	Imagine (Updated)

Your project

By the end of Module 2 you will:

1. Choose your own listable population and one variable to summarize with a proportion or mean.
2. Draw a simple random sample without replacement, check conditions, and compute the correct 95% CI (including FPC when appropriate).
3. Submit a one-page summary and runnable R notebook with your code.
   ::: {.callout-tip icon=false} ## Check yourself — Module 2 preview
   1. In one sentence each: what is a **parameter**? What is a **statistic**?
    2. Which symbol is typically Greek for a population mean? Which letter denotes the sample mean?
    3. You measure all $N = 50$ courses in a department. Do you need a confidence interval for the proportion with a lab? Why or why not?
    4. Project 2 asks for a 95% CI. Name the five steps of the statistical process in order.

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Answers — Module 2 preview

1. **Parameter:** a number describing the entire population (usually unknown). **Statistic:** a number computed from the data you observed.
2. μ (parameter); $\bar{x}$ (statistic).
3. **No**, if all units are observed that is a **census** and we can just compute the population proportion exactly (no need for a good guess)
4. Ask → Plan → Collect → Analyze → Conclude.

Check yourself — Module 2 preview

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Answers — Module 2 preview

1. **Parameter:** a number describing the **population** (usually unknown). **Statistic:** a number computed from the **sample** you observed.
2. μ (parameter); $\bar{x}$ (statistic).
3. **No** — that is a **census**; you know the population proportion exactly (no sampling error for that parameter).
4. Ask → Plan → Collect → Analyze → Conclude.